CSE373, MAT373: Analysis of Algorithms, Spring 2024 **Date:** March 19, 2024

Midterm Exam

(5:35 PM – 6:50 PM : 75 Minutes)

- This exam will account for 25% of your overall grade.
- There are five (5) questions, worth 75 points in total. Please answer all of them in the spaces provided.
- There are 18 pages including 9 blank pages. Please use the blank pages if you need additional space for your answers.
- This is a *closed book*, *closed notes* exam. *No cheat sheets* are allowed.
- You are *not allowed to use calculators.*

Good Luck!

Question	Pages	Parts	Points	Score
1. Recurrences	$2-4$	$(a)-(d)$	$4\times 4=16$	
2. Asymptotic Bounds	$5-7$	$(a)-(d)$	$4\times 4=16$	
3. Yet Another Sum	$8-9$	–	8	
4. Recursive Divide and Conquer	$10-14$	$(a)-(b)$	$7+8=15$	
5. Quicksort	$15-18$	$(a)-(d)$	$4\times 5=20$	
Total	–	–	75	

Name: ______________________________

SBU ID: ______________________________

Question 1. [16 Points] Recurrences. For each of the functions below either solve it using the Master Theorem (and state the case that applies) or state why the Master Theorem does not apply. Assume that all log's are of base 2 and $T(n) = \Theta(1)$ for $n \leq 1$.

(a) [**4 Points**] $T(n) = 4 \cdot T\left(\frac{n}{n^{\frac{1}{\log n}}}\right) + n$

(b) [**4 Points**] $T(n) = \left(\sum_{i=1}^{\log n} T\left(\frac{n}{2^i}\right)\right) + 2n$, where n is a power of 2

(c) [**4 Points**] $T(n) = 3 \cdot T\left(\frac{n}{2}\right) - n^2$

(d) [**4 Points**] $T(n) = \frac{1}{3} \cdot T(2n) - n^2$

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Question 2. [**16 Points**] **Asymptotic Bounds.** For each of the following statements, state whether it is *True* or *False*. Justify each answer. Assume that all log's are of base 2.

(a) [**4 Points**] $\frac{1}{n} = \mathcal{O}\left(\frac{1}{n^2}\right)$

(b) [**4 Points**] $\log(n!) = \mathcal{O}(n \log n)$

(c) [**4 Points**] $n^{\log \log n} = \omega\left((\log n)^{\log n}\right)$

(d) [**4 Points**] $\sum_{i=0}^{n} \frac{1}{2^i} = \Theta\left(\sum_{i=0}^{n} \frac{1}{n^i}\right)$

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Question 3. [8 Points] Yet Another Sum. Give the best upper bound you can derive on the worst-case running time of SUM-5 for an input of size n. You must show your analysis.

```
SUM-5( n, A[ 1 : n ] )
s ← 0
d ← 1
x ← log_e log_e n
for i ← 1 to n do
    d ← d × (i/x)
    j ← ⌈d⌉
    while j ≤ n do
        s ← s + A[j]
        j ← j + ⌈d⌉
return s
```

You may find the following series useful (which holds for all real values of x):

$$e^x = \sum_{i=1}^{\infty} \frac{x^i}{i!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad \dots \quad \dots,$$

where $e \approx 2.7182818$ is the base of natural logarithm.

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Question 4. **[15 Points] Recursive Divide and Conquer.** The Rec-Inplace-FFT function given below is an in-place version of the Fast Fourier Transform (FFT) algorithm. It is based on recursive divide and conquer and uses the Unshuffle function (also given below) as a subroutine. The Unshuffle function, too, uses recursive divide and conquer.

Rec-Inplace-FFT(A, n)

Input: An array $A[1:n]$ of n complex numbers, where $n = 2^k$ for some integer $k \geq 0$.

Output: The Discrete Fourier Transform (DFT) of the input array computed in place in $A[1:n]$.

Algorithm:

if $n > 1$ **then**
 Unshuffle(A, n)
 Let A_L denote $A\left[1 : \frac{n}{2}\right]$ (i.e., the left half of A) and let A_R denote $A\left[\frac{n}{2}+1 : n\right]$ (i.e., the right half of A)
 Rec-Inplace-FFT(A_L, $\frac{n}{2}$)
 Rec-Inplace-FFT(A_R, $\frac{n}{2}$)
 $y \leftarrow \cos\left(\frac{2\pi}{n}\right) + i \cdot \sin\left(\frac{2\pi}{n}\right)$ $\{i = \sqrt{-1}\}$
 $z \leftarrow 1$
 for $j \leftarrow 1$ **to** $\frac{n}{2}$ **do**
  $l \leftarrow A_L[j]$
  $r \leftarrow A_R[j]$
  $A_L[j] \leftarrow l + z \cdot r$
  $A_R[j] \leftarrow l - z \cdot r$
  $z \leftarrow z \cdot y$

Unshuffle(A, n)

Input: An array $A[1:n]$ of n complex numbers, where $n = 2^k$ for some integer $k \geq 0$.

Output: The entries of $A[1:n]$ rearranged in place so that all entries located at odd (resp. even) indices of the input array are moved to the left (resp. right) half of the array without changing the order in which they appeared in the input array.

Algorithm:

if $n > 2$ **then**
 Let A_L denote $A\left[1 : \frac{n}{2}\right]$ (i.e., the left half of A) and let A_R denote $A\left[\frac{n}{2}+1 : n\right]$ (i.e., the right half of A)
 Unshuffle(A_L, $\frac{n}{2}$)
 Unshuffle(A_R, $\frac{n}{2}$)
 for $j \leftarrow 1$ **to** $\frac{n}{4}$ **do**
  $A_L\left[\frac{n}{4}+j\right] \leftrightarrow A_R[j]$ $\{swap\}$

(a) [**7 Points**] Let $T'(n)$ denote the worst-case running time of UNSHUFFLE on an array of size n. What is the cost of the **divide** step of the algorithm? What is the cost of the **combine** step? Write a recurrence relation for $T'(n)$ and solve it.

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(*b*) [**8 Points**] Let $T(n)$ be the worst-case running time of REC-INPLACE-FFT on an array of size n. What is the cost of the **divide** step of the algorithm? What is the cost of the **combine** step? Write a recurrence relation for $T(n)$ and solve it.

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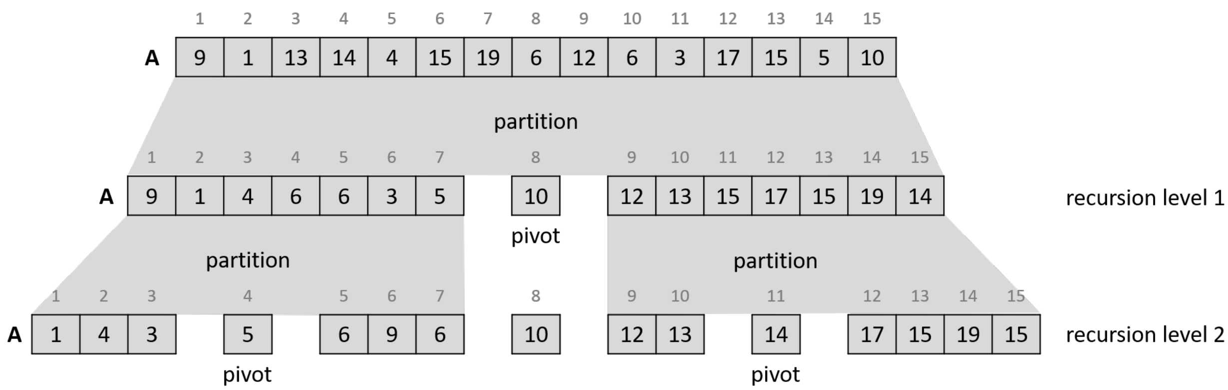
Question 5. [20 Points] Quicksort.

Recall the in-place partitioning algorithm we learned in the class which is used by quicksort to partition the numbers in the array or subarray it is working on around the pivot. We always chose the last element of the given array/subarray as the pivot.

Given an array $A[1:15]$ as input, the following example shows the state of A after partitioning in the first two levels of recursion of quicksort.



You can describe the state of A after partitioning in recursion level 1 as:

$$[\ 9,\ 1,\ 4,\ 6,\ 6,\ 3,\ 5\ ]\ \ 10\ \ [\ 12,\ 13,\ 15,\ 17,\ 15,\ 19,\ 14\ ]$$

Then after partitioning in recursion level 2 as:

$$[\ [\ 1,\ 4,\ 3\ ]\ \ 5\ \ [\ 6,\ 9,\ 6\ ]\ ]\ \ 10\ \ [\ [\ 12,\ 13\ ]\ \ 14\ \ [\ 17,\ 15,\ 19,\ 15\ ]\ ]$$

Now, given the following array $A[1:16]$ of numbers, you will have to describe its state after partitioning at recursion levels 1, 2, 3 and 4.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	97	22	31	38	4	71	55	12	64	88	33	17	48	92	8	43

(*a*) [**5 Points**] Describe the state of A after partitioning at recursion level 1.

(*b*) [**5 Points**] Describe the state of A after partitioning at recursion level 2.

(*c*) [**5 Points**] Describe the state of A after partitioning at recursion level 3.

(*d*) [**5 Points**] Describe the state of A after partitioning at recursion level 4.

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